

LEARNING CHESS WITHOUT CHESS KNOWLEDGE: SUPERVISED BOOTSTRAPPING AND GATED SELF-PLAY REINFORCEMENT LEARNING

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ABSTRACT

This report documents a project on training a compact neural network to play chess without hand-coded chess knowledge, using the AlphaZero recipe: a residual policy–value network trained first by supervised imitation of human games and then improved by self-play guided by Monte Carlo Tree Search. A 760 717-parameter network was trained on 676 648 positions mined from the Lichess open database, reaching a policy cross-entropy of 2.06. The central finding is methodological rather than architectural. The evaluation procedure originally used scored games from White’s perspective while the network alternated colours, which rendered 159 logged evaluation points across two training runs statistically vacuous. Once corrected scoring was applied, direct head-to-head play revealed that 300 iterations of self-play had made the model *weaker* than its own starting point (2.5 to 5.5 over eight games) even as training loss fell monotonically. Two root causes were identified and fixed: policy targets formed from 40-visit search histograms are dominated by sampling noise, and a mirror data-augmentation step was generating physically impossible board states. A corrected pipeline adds acceptance gating: over 199 further iterations the gated model measured 46.2% against its frozen starting point (95% CI 37.4–55.1, indistinguishable) while the candidate stream it was trained from averaged 42.2% ($p = 0.004$). Gating prevented the regression without producing improvement; the residual barrier is sample efficiency.

Index Terms— Reinforcement Learning, Monte Carlo Tree Search, Deep Learning, Self-Play, Evaluation Methodology, Experimental Design

1. INTRODUCTION

Game playing has long served as a testbed for machine learning because the reward is unambiguous and data can be generated without human annotation. AlphaZero [1] showed that a single neural network, trained purely by self-play and guided by Monte Carlo Tree Search (MCTS), can reach superhuman strength in chess, shogi and Go without any domain-specific heuristics. The recipe is attractive from a data-science standpoint because it is a closed loop: the model generates its own training distribution, and improvements compound.

That closed loop is also its central risk. When a model manufactures its own labels there is no held-out set whose labels are known to be correct, and the quantity one cares about — playing strength — is not the quantity being minimised. The two can move in opposite directions with no warning in the loss curve.

This project reproduces the AlphaZero recipe at a scale that fits on a single desktop machine, and then subjects it to the measurement discipline that self-play demands. Section 2 states the problem and objectives. Section 3 describes the data pipeline, the representation, the network and the two training stages. Section 4 presents the experimental results, including a defect in the evaluation methodology that invalidated an entire training run. Section 5 interprets these findings, and Section 6 concludes.

2. PROBLEM STATEMENT AND OBJECTIVES

2.1. Problem Statement

Given only the rules of chess and a corpus of human games, train a neural network with fewer than one million parameters to select strong moves, subject to the constraint that all data generation, training and evaluation run on a single consumer machine (six CPU cores, one mid-range GPU) within a few hours per experiment. The network receives no evaluation function, no opening book, no endgame tablebase and no hand-engineered features; it must learn the value of a position and the plausibility of a move entirely from data.

The scale constraint is not incidental. AlphaZero used 5000 TPUs and 800 simulations per move. The question here is therefore not whether the recipe reaches grandmaster strength, but *which parts of it survive aggressive downscaling* — and how one would know.

2.2. Objectives

1. To design and implement an end-to-end pipeline: data acquisition, board representation, network, search, training and evaluation.
2. To bootstrap the network by supervised imitation of human play, and quantify the resulting move-prediction quality.
3. To improve the bootstrapped network by self-play reinforcement learning, and *verify* that improvement with a metric valid under self-play conditions.
4. To diagnose any failure to improve, and to make the training procedure robust against it.

Objective 3 is deliberately phrased to separate *running* the algorithm from *demonstrating* that it worked. That separation turned out to carry the entire result.

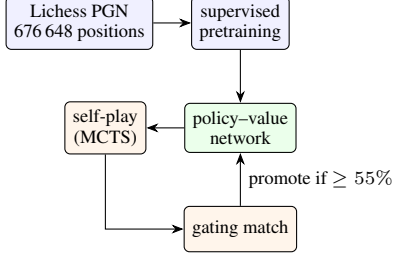


Fig. 1. The two-stage pipeline. Supervised pretraining supplies the initial policy; self-play then proposes updates, which are accepted only if they win a head-to-head match against the incumbent weights.

3. METHODOLOGY

3.1. Overall Workflow

Training proceeds in two stages (Figure 1). Stage 1 fits the network to human moves, giving it an immediate grasp of openings and basic tactics. Stage 2 replaces the human teacher with the network’s own search. An acceptance gate, added after the diagnosis in Section 4.4, sits between the two.

3.2. Data Pipeline

Data come from the Lichess open database [2], distributed as Zstandard-compressed PGN. Monthly dumps are streamed and decompressed incrementally rather than materialised, since a single month exceeds available memory when expanded.

Quality filtering is applied at the game level: only games in which *both* players are rated at least 1750 Elo are retained. This admits roughly 12% of games, giving a corpus of 676 648 positions from about 9600 games out of the 80 000 read.

Each retained position yields one training example: an $18 \times 8 \times 8$ tensor, the index of the move the human actually played, and the game outcome expressed from the side to move. A subset of 40 000 examples is frozen as a *rehearsal anchor* (Section 3.6), and 6000 positions taken five to eight plies into games are retained as a self-play opening curriculum.

3.3. Representation and Action Encoding

The board is encoded as 18 binary planes of 8×8 : twelve piece-placement planes (six piece types $\times$ two colours), four castling-rights planes, one en-passant plane and one side-to-move plane.

The action space follows the Leela Chess Zero scheme [3]: each of the 64 origin squares carries 73 move planes — 56 queen-style moves (eight directions $\times$ seven distances), eight knight moves and nine underpromotions — giving $64 \times 73 = 4672$ actions, indexed as $\text{index} = \text{from} \times 73 + \text{plane}$. A round-trip self-test verifies that every legal move in a sample of positions encodes and decodes without collision.

3.4. Network Architecture

The network is a residual convolutional trunk [4] with two heads: a 3×3 convolution mapping 18 input planes to 64 channels, ten residual blocks of two 3×3 convolutions with batch normalisation [5], a policy head (1×1 convolution to 73 planes, reshaped to 4672 logits)

and a value head (1×1 convolution to one channel, a fully connected layer to 64 units, and a scalar through tanh). Total trainable parameters: 760 717.

The heads share the trunk, so a single representation must serve both *who is winning* and *which moves are plausible*.

3.5. Stage 1: Supervised Pretraining

Given a position s , the human move m and the outcome $z \in \{-1, 0, +1\}$, the loss is

$$\mathcal{L}_{\text{sup}} = -\log p_{\theta}(m | s) + (v_{\theta}(s) - z)^2, \quad (1)$$

optimised with Adam [6], a cosine learning-rate schedule from 10^{-3} to 10^{-4} , and batch size 256 for three epochs.

3.6. Stage 2: Self-Play Reinforcement Learning

The network plays itself. Each move is selected by MCTS [7, 8], which descends the tree by the PUCT rule [9]

$$U(s, a) = Q(s, a) + c_{\text{puct}} P(s, a) \frac{\sqrt{N(s)}}{1 + N(s, a)}, \quad (2)$$

where Q is the child’s mean value, P the network prior and N the visit count. Leaves are evaluated by the network, terminal positions by the true result, and Dirichlet noise is added at the root during self-play only. The normalised visit distribution π becomes the policy target and the result z the value target, giving the AlphaZero loss

$$\mathcal{L} = \mathbb{E}_s \left[-\pi^{\top} \log p_{\theta}(s) + (v_{\theta}(s) - z)^2 \right] + \lambda \|\theta\|^2. \quad (3)$$

Two additions guard against catastrophic forgetting [10], the first being rehearsal on a frozen exemplar set [11]: 30% of every mini-batch comes from the anchor, and self-play games start from curriculum positions 85% of the time. Self-play runs in six CPU workers overlapped with GPU training, so self-play weights lag one iteration — a standard relaxation.

3.7. Evaluation Methodology

Because self-play provides no held-out labels, playing strength must be measured by playing games. Three properties are required for such a measurement to carry information:

1. **Correct attribution.** Score from the *network’s* perspective: if it alternates colours and results are tallied as raw PGN strings, every win as Black counts as a loss.
2. **Independent games.** Two deterministic players replay one identical game, so an N -game match carries one game of information unless openings are randomised.
3. **An informative opponent.** A frozen reference measures progress; one that improves alongside the model measures nothing.

All three were violated originally; their restoration is what makes Section 4 interpretable.

4. EXPERIMENTS AND RESULTS

4.1. Experimental Setup

All experiments ran on an Intel i5-12400F (6C/12T) with an NVIDIA RTX 3060, in PyTorch 2.13 [12] with `python-chess` [13].

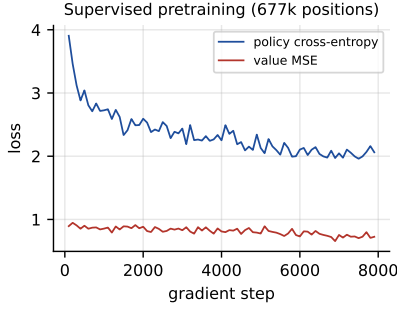


Fig. 2. Supervised stage. Policy cross-entropy converges to ≈ 2.06 ; value MSE plateaus near 0.73, reflecting the intrinsic difficulty of predicting a game outcome from a single mid-game position.

Self-play is CPU-bound, the GPU serving only batched gradient steps, so low GPU utilisation is expected. Throughput at twelve games per iteration on six workers is 42 s at 40 simulations, 180 s at 128.

4.2. Supervised Pretraining

Policy cross-entropy fell from 3.90 to 2.06 over three epochs (Figure 2) — roughly 13% probability on the move a strong human actually chose, against a uniform-over-legal-moves baseline near 3.4.

Behavioural probes confirm this reflects real competence: the raw policy assigns 0.594 to e4 and 0.313 to d4, answers e4 with e5 or c5, captures a hanging queen and finds mate in one, scoring 85% against a random opponent.

4.3. A Defect in the Evaluation Metric

The evaluation harness alternated the network between colours but scored wins-as-White plus half the draws. Combined with the absence of opening randomisation, the consequence is a tally that can only be symmetric, and a score pinned near 50% regardless of the model:

```
[iter 454] eval: {1-0: 0, 0-1: 0, 1/2-1/2: 10} (score 50%)
[iter 499] eval: {1-0: 0, 0-1: 0, 1/2-1/2: 10} (score 50%)
```

159 evaluation points across two runs conveyed no information. The same defect reported the pretrained model at 44% against a random opponent when its true score is 85%.

4.4. Self-Play Regressed

With corrected scoring and randomised openings, the final model of the 300-iteration self-play run was played directly against its own starting point over eight games:

v5 iter_500 2.5 – 5.5 pretrained iter_200

Self-play made the model *worse*. Throughout those 300 iterations the training loss fell monotonically from 2.36 to 1.85 (Figure 3). Loss is not a strength signal.

An earlier run without the rehearsal anchor failed more visibly: its policy degenerated to opening with f3, its value head returned -0.270 for the balanced initial position, and it no longer captured a hanging queen — textbook catastrophic forgetting [10]. Rehearsal prevented that collapse, but preventing collapse is not improvement.

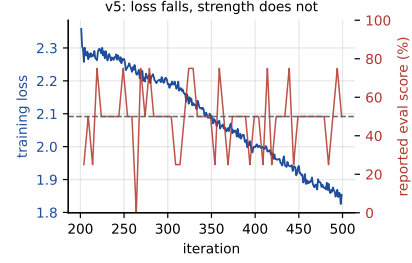


Fig. 3. The v5 run. Training loss (left axis) falls throughout, while the reported evaluation score (right axis) is pinned at 50% by the defect of Section 4.3. Direct head-to-head play shows the model in fact got weaker.

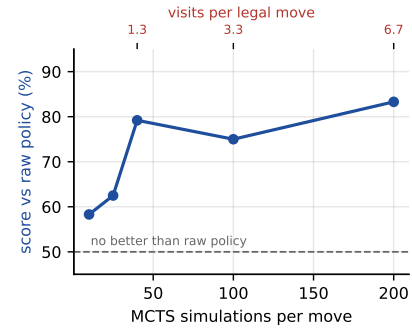


Fig. 4. MCTS against the network's own greedy policy. Search is clearly stronger than the raw policy even at 40 simulations, so the *argmax* of the search is a good move; the upper axis shows why the *distribution* is nonetheless a poor target.

4.5. Diagnosis: Search Quality Versus Target Quality

A natural hypothesis is that search is too weak to teach the network. It was tested by playing MCTS against the same network's raw policy head at varying simulation counts (Figure 4). Search wins comfortably — 79% at only 40 simulations — so the hypothesis is false.

The distinction matters. Equation (3) trains on the full visit distribution π , not on its *argmax*. With 40 simulations spread over roughly 30 legal moves, π is a histogram of about 1.3 visits per move: its mode is informative, its shape is sampling noise. Fitting a sharp supervised policy to that noise flattens it.

A second, independent defect compounded this. A mirror augmentation flipped board files on half of every minibatch but did not exchange the kingside and queenside castling planes. After a file flip the king stands on d1 while the kingside-castling plane remains set — a position that cannot arise in chess, paired with features contradicting the pieces. Chess is not mirror-symmetric, precisely because castling and pawn direction break the symmetry; roughly half of every batch, including the rehearsal slice, was corrupted.

4.6. Corrected Pipeline

Four changes: castling-aware augmentation (positions retaining castling rights are not mirrored), 128 simulations instead of 40 (≈ 4.3 visits per legal move), 80 gradient steps instead of 200, and a lower learning rate.

The decisive change is *acceptance gating* [14]: self-play always

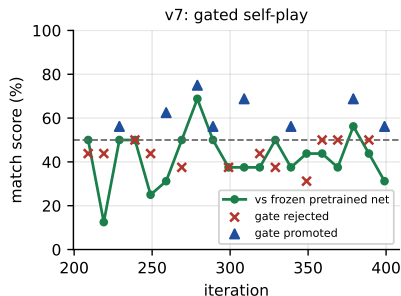


Fig. 5. The corrected run. Each gate point plots the candidate’s score against the incumbent (markers) and against the frozen pretrained network (line). The dashed line is equal strength.

Table 1. Corrected run. Head-to-head figures are 40-game matches with randomised openings; intervals are 95% CIs. Training loss fell 2.560→2.473 over 199 iterations.

Quantity	Value
Gate promotions (noise predicts 7.8)	8 / 20
Candidate vs. baseline (20 gates)	42.2% ($p = 0.004$)
Gated model vs. baseline	46.2% [37.4, 55.1]
Gated model vs. random	86.2% [79.2, 93.3]
Baseline vs. random	87.5% [80.7, 94.3]

generates from the best weights so far, and updated weights reach the deployed model only after scoring at least 55% against the incumbent.

The run completed 199 iterations, settled by a 40-game match (random openings, alternating colours) against the frozen network; at 40 games the standard error is 7.9 points, against 17.7 for the eight-game training matches (Table 1).

It scored 46.2% (95% CI 37.4–55.1, $z = -0.83$): no detectable difference from its starting point, both models beating a random mover identically (86.2% and 87.5%). Self-play produced no improvement — but nor the regression of Section 4.4.

The mechanism is visible in the intermediate data: the *candidate* network, trained continuously and repeatedly rejected, averaged 42.2% over 20 measurements ($z = -2.87$, $p = 0.004$). Self-play was still degrading the network, and the gate kept that damage out of the deployed weights.

The guarantee is weaker than it appears: 8 of 20 gates promoted against 7.8 expected under the null (Section 5).

5. DISCUSSION

Optimisation success is not task success. A falling loss curve coexisted with a model getting worse at the task. Equation (3) is computed against targets the model itself produced, so minimising it measures self-agreement, not chess strength. Every self-training pipeline shares this structure and this failure mode.

A broken metric is worse than no metric. The defect was costly not because it was subtle but because it was *reassuring*: it returned 50%, a plausible constant, for 159 consecutive measurements across two multi-hour runs. The symmetric tallies were visible in every log line and read as evenly matched play rather than as the signature of a bug.

Sample efficiency governs downscaling. The failure was not a shortage of compute but its allocation: cutting simulations from 800 to 40 preserves the argmax while destroying the calibration of the distribution, and Equation (3) consumes the distribution. When downscaling, the question is not what a parameter does at full scale but which statistic of it the loss actually reads.

A safeguard is only as good as its test. Gating worked in aggregate — rejected candidates were significantly worse than baseline, the protected model was not — but individual decisions were indistinguishable from chance, an eight-game match resolving nothing at this effect size. Having diagnosed one metric that measured nothing, we built a safeguard and assumed it worked: the same error one level up.

Limitations. The value head plateaus at MSE 0.73, and absolute strength is uncalibrated against a rated engine, so all results are relative.

6. CONCLUSION AND FUTURE WORK

Supervised pretraining on 676648 Elo-filtered Lichess positions gave a 760717-parameter network that plays principled openings, wins material and finds simple mates, scoring 85% against a random opponent.

The substantive contribution is negative and methodological. Self-play as originally implemented degraded the model, and the evaluation procedure could not detect it. Correcting the metric exposed the regression; controlled experiments isolated one statistical cause (visit histograms too sparse to be distributional targets) and one representational cause (an augmentation producing illegal boards); and gating held the deployed model level (46.2%, CI 37.4–55.1) while rejected candidates drifted significantly below baseline. Self-play at this scale neither helped nor, once gated, hurt. The binding constraint is sample efficiency: 128 simulations give 4.3 visits per legal move against roughly 27 for AlphaZero.

Future work follows from the diagnosis: batched leaf evaluation and subtree reuse would raise simulations several-fold at equal wall-clock cost; sequential-probability-ratio testing [15] would set gate decisions at a stated confidence level; and calibration against Stockfish at reduced strength would yield an Elo estimate [16].

7. ARTIFACTS AND DEMONSTRATIONS

- **Code:** <https://github.com/realgaauravvyas/chess-ai> — network, search, both training stages, the dashboard, and the scripts reproducing every number in this report. `RESULTS.md` records the findings and the defects fixed.
- **Live demo:** <https://realgaauravvyas.github.io/chess-ai/> — play the trained network in a browser, with the search’s principal variation, its candidate moves and the training curves.
- **Video:** [URL TO BE ADDED]

Disclaimer. AI-assisted tools were used during the development of this project, for code review, debugging and drafting. All experimental results reported here were produced by the author’s own runs and are reproducible from the linked repository.

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